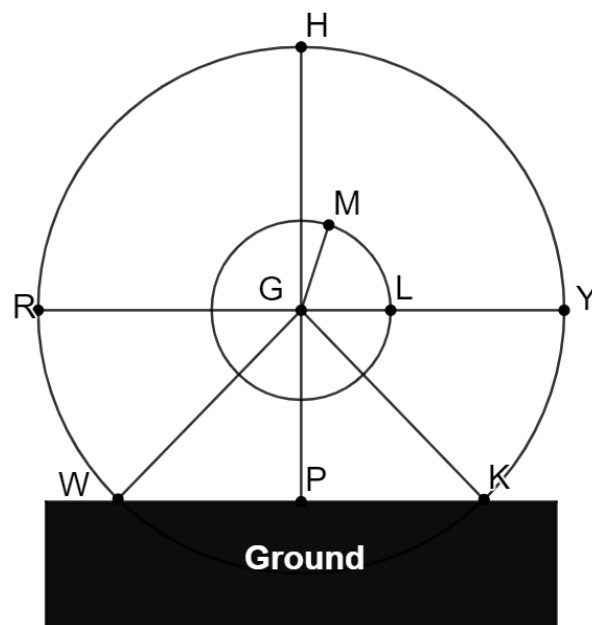


1. Guangzhou Circle is a circular building in Guangzhou, China designed by Italian architect Joseph di Pasquale. The building's height is 138 meters (m) and it has a diameter of 194 m. Guangzhou Circle has a circular hole at its center that is open with views of the Zhujiang River. The diameter of the circular hole is 48 m. A diagram of the building is shown where the smaller circle represents the circular hole, $\overline{GL}$ is the radius of the circular hole, $\overline{HP}$ is height of the building, and $\overline{RY}$ is diameter of the building.

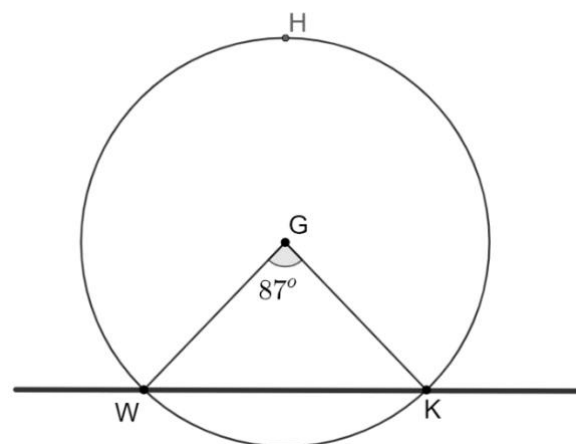


- a. Determine the area, in square meters, of the sector $\widehat{LM}$ if $m\angle LGM = 72^\circ$. Round your answer to the nearest thousandth.

area	
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- b. The part of the building that rests on the ground is not circular. In the diagram circle G represents the Guangzhou circle and $\overline{WK}$ represents the ground.

Determine the length of $\widehat{WK}$ if $m\angle W GK = 87^\circ$. Round your answer to the nearest thousandth meter.

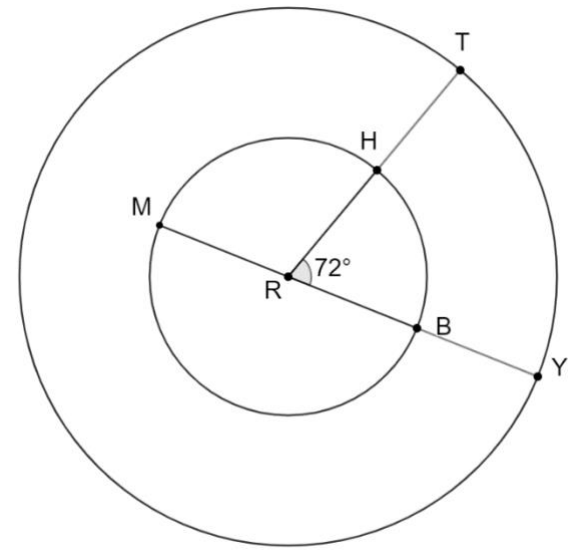


arc length	
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2. Two circles are shown where $RB = 27$ centimeters (cm), $HT = 25.5$ cm, $m\angle HRB = 72^\circ$.

- a. Determine the length of $\widehat{MH}$. Leave in terms of π .

arc length	
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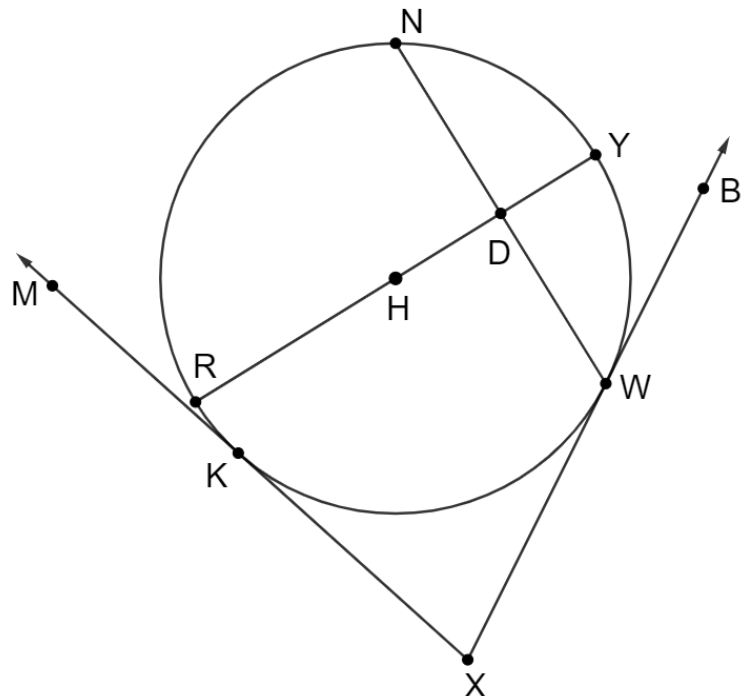


- b. Determine the length of $\widehat{TY}$. Leave in terms of π .

arc length	
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3. Circle H is shown where points N , Y , W and K are on the circle, $m\widehat{NY} = 58^\circ$, $m\widehat{RK} = 16^\circ$, $m\widehat{KW} = 106^\circ$, and $\overline{RY}$ is a diameter and a perpendicular bisector of $\overline{NW}$.

Determine $m\angle KXW$.



$m\angle KXW =$	
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6. Circle N is shown where $\triangle LNW$ is isosceles, $m\widehat{LH} = 105^\circ$, $m\angle LWN = 25^\circ$, and $m\angle DNW = 50^\circ$.

Determine each measure.

a. $m\angle LBH$

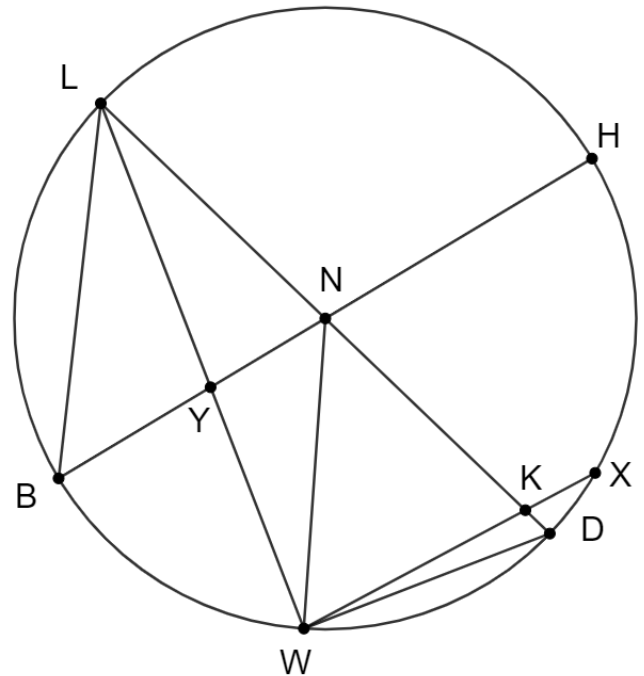
$m\angle LBH =$	
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b. $m\angle DWL$

$m\angle DWL =$	
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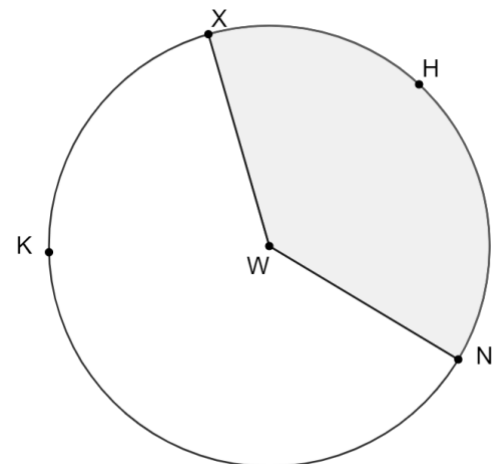
c. $m\widehat{BW}$

$m\widehat{BW} =$	
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7. Circle W is shown where $m\widehat{XHN} = 137^\circ$ and $WN = 5.8$ centimeters. Determine the area of the shaded sector. Round your answer to the nearest thousandth.

area	
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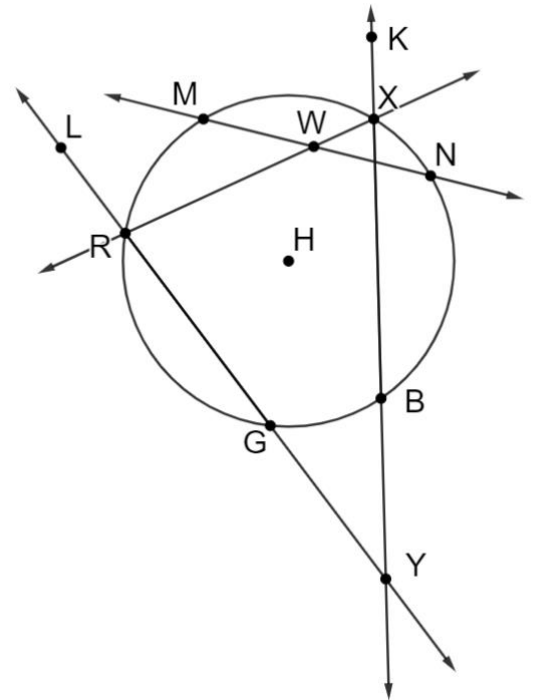
8. Circle H is shown where points $M, X, N, B, G,$ and R are on the circle and $m\widehat{RM} = (9x - 4)^\circ$, $m\widehat{MX} = (8x + 14)^\circ$, $m\widehat{NX} = (5x - 2)^\circ$, $m\widehat{GB} = 40^\circ$, and $m\angle GYB = (7x - 6)^\circ$.

- a. Determine the value of x .

$x =$	
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- b. Determine $m\angle XWN$.

$m\angle XWN =$	
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9. Circle M is shown where points $X, C,$ and B are on the circle and $m\angle XMC = (4y - 16)^\circ$, $m\angle BMC = (3y + 6)^\circ$, and $m\angle XMB = (5y + 10)^\circ$.

- a. Determine the value of y .

$y =$	
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- b. Determine $m\angle KXC$.

$m\angle KXC =$	
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